

Geometry

Unit 9

Lesson 7 Homework

Name_____ **Answer Key** _____

Date_____

Period_____

1. Oliver lives by the highway. He noticed several tractor trailers drive by every afternoon. Many of the tractor trailers are tankers. The cylinder on a tanker can be 42.6 feet long with a radius of 8.3 feet. What is the maximum volume of the cylinder on a tanker?



$$V = \pi(8.3)^2(42.6) = 2934.714\pi \approx 9219.68 \text{ ft}^3 \text{ (when using the pi button)}$$

2. Oliver counted 16 tanker trucks one Saturday. If every tanker truck was full, what was the total volume of cargo that passed by that day?

$$9219.68(16) = 147,514.88 \text{ ft}^3$$

3. A baseball has a diameter of 7.5 centimeters. What is the volume of a baseball? Round to the nearest hundredth.



$$V = \frac{4}{3}\pi(3.75)^3$$

$$V \approx 220.89 \text{ cm}^3 \text{ (when using the pi button)}$$

4. The diameter of a softball is 8.89 centimeters. What is the volume of a softball? Round to the nearest hundredth.

$$V = \frac{4}{3}\pi(4.445)^3 \approx 367.88 \text{ cm}^3 \text{ (when using the pi button)}$$

5. What is the difference in volume between a baseball and a softball?

$$\approx 146.99 \text{ cm}^3$$

6. Penelope went on a road trip. During the drive, she noticed that some hay bales were rectangular prisms and some were cylindrical. She learned that the first hay bale she saw that was a rectangular prism was 16 inches high by 22 inches wide by 44 inches long. What was the volume of the first hay bale? Round to the nearest hundredth.

$$V = 16(22)(44) = 15,488 \text{ in}^3$$

7. The second hay bale was cylindrical with a 5 foot diameter and a 4 foot width. What was the volume of the second hay bale, in cubic inches? Round to the nearest hundredth.

$$V = \pi(2.5)^2(4) \text{ (feet)}$$

$$V = \pi(30)^2(48) \text{ (inches)}$$

$$\approx 135,716.80 \text{ in}^3 \text{ (when using the pi button)}$$

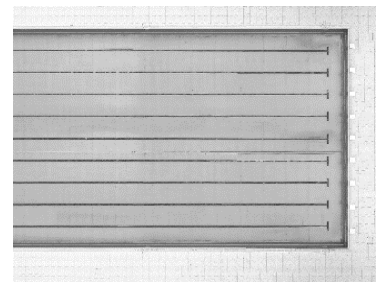


8. Which hay bale is larger?

The cylindrical bale of hay is larger.

9. The high school swimming pool is 25 meters long, 15 meters wide, and 3 meters deep. What is the volume of water in a full pool?

$$V = 25(15)(3) = 1125 \text{ m}^3$$



10. If the pool loses $\frac{1}{10}$ of its water depth during a butterfly swim event, what is the new volume of water?

$$\frac{9}{10}(1125) = 1012.5 \text{ m}^3$$